



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 31



HANDY Shredder
Model: Saber 13.5 Hp

HANDY SABER 13.5 HP SHREDDER

Test Report No.	:	2013 - 31
Date of Test	:	February 28, 2013
Test Valid Until	:	February 28, 2018
Brand	:	HANDY
Model	:	Saber 13.5 Hp
Type	:	Vertical single shaft, wheelbarrow-type
Manufacturer	:	HANDY 4-IN-1 INC. 116 4 th St., Brgy. Mariana New Manila, Quezon City
Test Requested by	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The HANDY Saber 13.5 Hp Shredder was tested using chopped and split “buko” husks with shell with an average Moisture Content of 82.3%.

The HANDY Saber 13.5 Hp Shredder had an average Input Capacity of 1129 kg/h of “buko” husk with shell at blade shaft speed of 3346 rpm.

The Fuel Consumption Rate of the 10.1 kW TECUMSEH ENDURO XL gasoline engine was 2.10 L/h.

Three persons were needed to operate the machine.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance. The test was conducted in Brgy. Mariana, New Manila, Quezon City on February 28, 2013. The test was carried out to assess and verify the following:

1. Machine specifications
2. Performance
 - a. Input capacity
 - b. Speed of components
 - c. Noise emitted by the machine
 - d. Fuel Consumption
3. Labor requirement

OPERATOR'S MANUAL

No Operator's Manual was submitted.

METHOD OF TEST

The shredder was tested following the "PAES 245-2010, Agricultural Machinery – Biomass Shredders – Methods of Test".

DESCRIPTION OF THE MACHINE

The HANDY Saber 13.5 Hp as shown in **Figure 1**, was a vertical single shaft, wheelbarrow-type shredder. It was powered by a 10.1 kW TECUMSEH ENDURO XL gasoline engine which drove directly the machine's blade systems through a belt-pulley transmission system. The machine consisted of the hopper and the shredding/chipping chamber.

The hopper (**Fig. 2**) was where the materials to be shred were fed. It was funnel-shaped and was hinged to the shredding section. Material to be shred was fed into the hopper to a system of rotating blades and ejected at the front of the machine.

The shredding section (**Fig. 3**) consisted of one piece wing blade and four pieces of corrugated blades. This section was also provided with a stationary counter blade.

Below the shredding section was the chipping section (**Fig. 4**) for woods which consisted of a chipper plate and two pieces chipper blades. Wood to be chipped was fed to a 66 mm diameter opening (**Fig. 5**) which is provided at the side of the machine.

The machine was a wheelbarrow type provided with two front rubber wheels (**Fig. 6**) and handles (**Fig. 7**) used during transport.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. HANDY Shredder
Model: Saber 13.5 Hp



Figure 2. Hopper

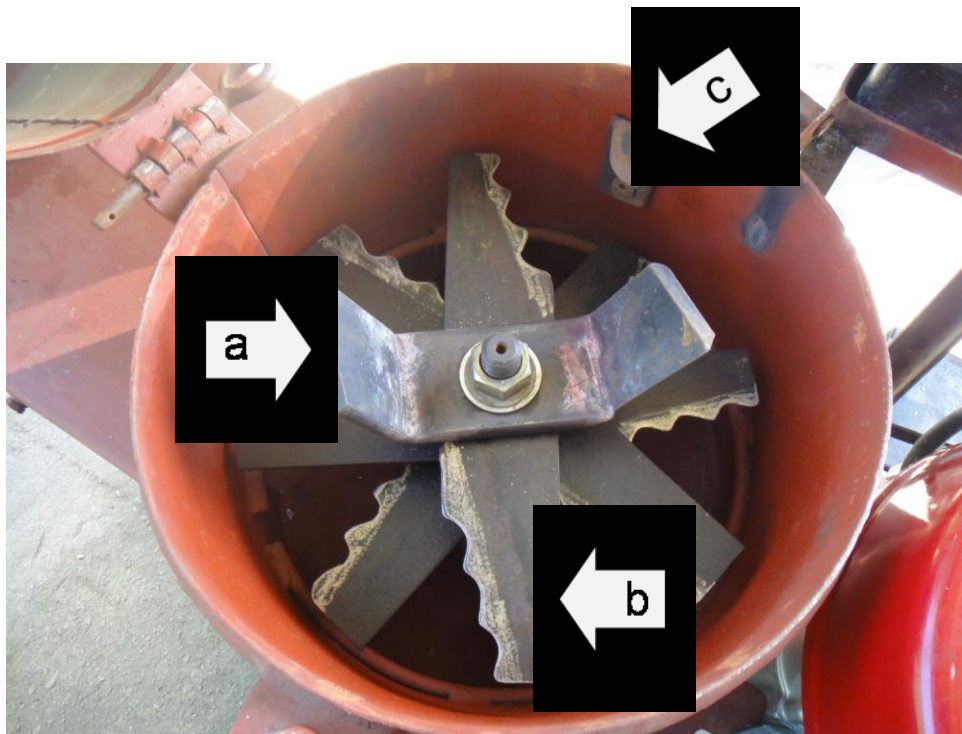


Figure 3. a) Wing blade
b) Corrugated blade
c) Counter blade

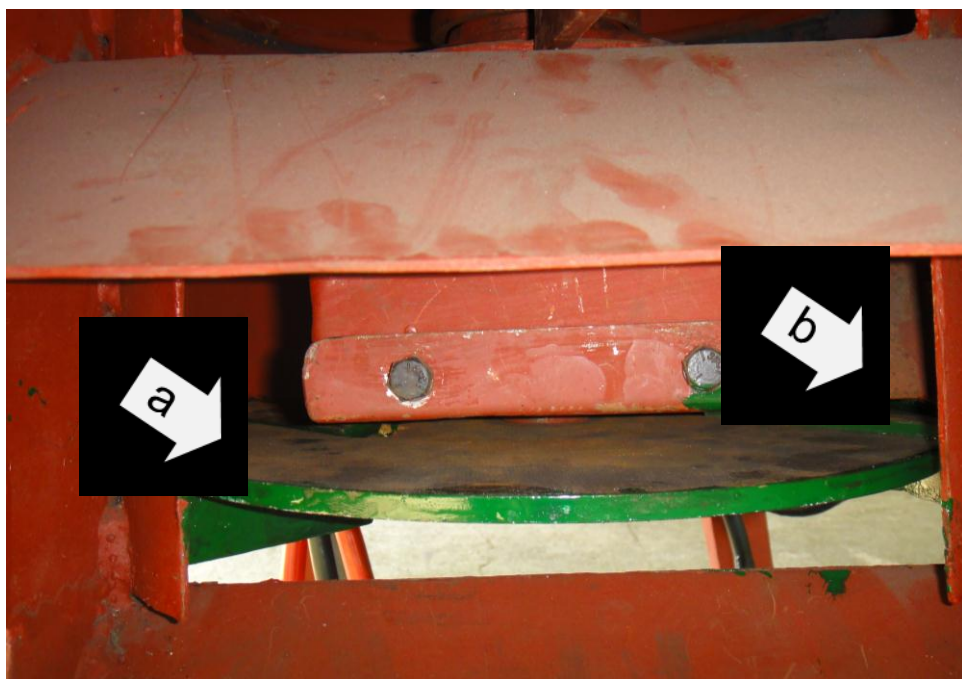


Figure 4. a) Chipping plate
b) Chipping blade

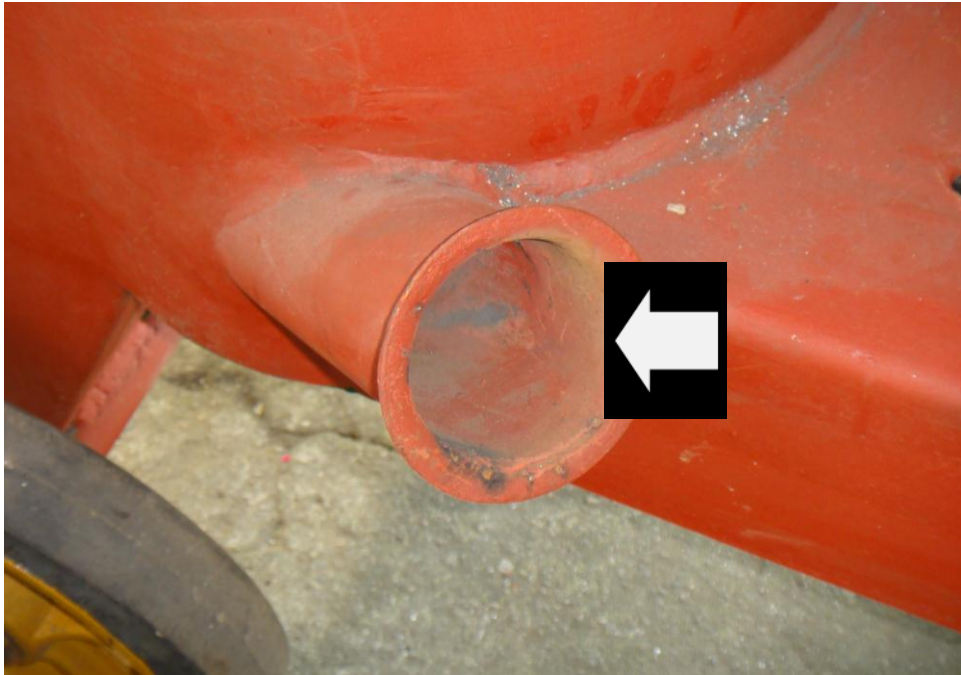


Figure 5. Chipping inlet opening



Figure 6. Rubber wheels

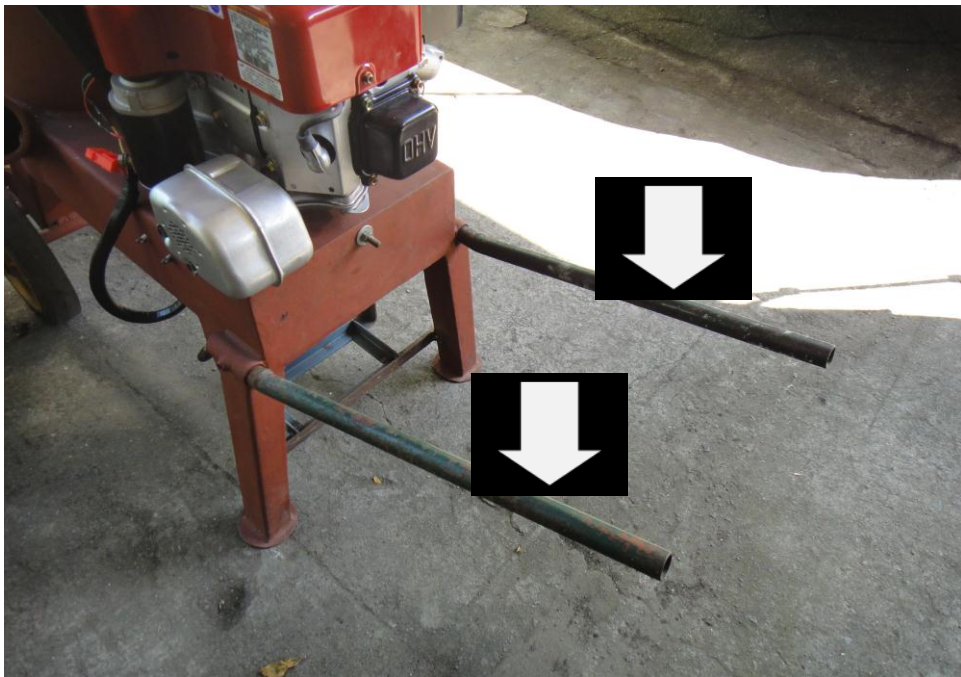


Figure 7. Handles

SPECIFICATIONS

Table 1. Detailed specifications of the HANDY Saber 13.5 Hp Shredder

A.	Brand	:	HANDY
B.	Model	:	Saber 13.5 Hp
C.	Serial No.	:	None
D.	Manufacturer	:	HANDY 4-IN-1 INC. 116 4 th St., Brgy. Mariana New Manila, Quezon City
E.	Overall Dimensions and Weight		
1.	Length (mm)	:	1070
2.	Width (mm)	:	585
3.	Height (mm)	:	1380
4.	Weight (kg) without engine	:	No Data
F.	Hopper		
1.	Type	:	Funnel shape
2.	Funnel bottom opening (mm)	:	550
3.	Material	:	Cold-Roll Sheet Ga. #16
G.	Shredding cylinder housing		
1.	Size, L x D (mm)	:	235 x 330
2.	Material	:	MS Plate 3 mm thick

Table 1. Detailed specifications of the HANDY Saber 13.5 Hp Shredder (cont'n.)

H.	Corrugated Blade		
1)	No. of pieces	:	4
2)	Size, L x W x T (mm)	:	310 x 65 x 6
3)	Material	:	Tool steel plate
I.	Wing Blade		
1.	No. of pieces	:	1
2.	Material	:	Spring steel 11 mm thick
J.	Counter Blade		
1.	No. of pieces	:	1
2.	Size, L x W x T (mm)	:	40 x 35 x 6
K.	Chipper Blade		
1.	No. of pieces	:	2
2.	Material	:	Bohler steel AISI-01 6 mm thick
L.	Chipper Plate		
1.	Material	:	MS Plate 8 mm thick
M.	Output chute		
1.	Size of opening, L x W (mm)	:	310 x 150
2.	Height from the ground (mm)	:	460

Table 1. Detailed specifications of the HANDY Saber 13.5 Hp Shredder (cont'n.)

N.	Engine Pulley	:	127 mm x 2B x 25 mm
O.	Shredding Shaft Pulley	:	127 mm x 2B x 25 mm
P.	Size of Belt	:	B - 40
Q.	Rubber Wheels		
	1. No. of pieces	:	2
	2. Size (D x W), mm	:	300 x 60
N.	Engine (Prime mover)		
	1. Brand	:	TECUMSEH
	2. Model	:	ENDURO XL
	3. Make	;	Canada
	4. Rated power (kW)	:	10.1
	5. Rated speed (rpm)	:	3600
	6. Type	:	Four-stroke, vertical crankshaft, OHV, single cylinder gasoline engine
	7. Starting system	:	Electric (battery)
	8. Cooling system	:	Air-cooled

RESULTS AND DISCUSSIONS

Results of performance tests are shown in **Table 2**.

Capacity

Test results showed that the machine had an Input Capacity of 1129 kg/h of buko husks with shell at blade shaft speed of 3346 rpm.

Fuel Consumption

The 101 kW TECUMSEH ENDURO XL gasoline engine had a Fuel Consumption Rate of 2.10 L/h.

Labor Requirement

Three persons were needed to operate the machine (**Fig. 8**).

Observations:

1. Mixture of split and chopped “buko” husks with shell (**Fig. 9**) was used in the test. Totally shredded “buko” husks (**Fig. 10**) were produced.



Figure 8. Operators of the machine



Figure 9. “Buko” husks with shell



Figure 10. Shredded coconut husks

PERFORMANCE TEST

Machine Tested : HANDY Shredder
Model: Saber 13.5 Hp

Date of Test : February 28, 2013

Location of Test : New Manila, Quezon City

Test Material Used : Mixture of split and chopped “buko” husks with shell

Table 2. Conditions of Test Materials

ITEMS	Data
1. Moisture Content, MC _{w.b.} (%)	82.3
2. Dimensions, L x W x T (mm)	159 x 129 x 40

Table 3. Result of Performance Test

ITEMS	Data
1. Speed of Components	
a) Engine (rpm)	
Without load	3618
With load	3380
b) Blade Assembly Shaft	
Without load	3621
With load	3346
2. Noise Level, dB(A)	
Without load	98.7
With load	101.0
3. Prime mover's Fuel Consumption (10.1 kW TECUMSEH ENDURO XL Gasoline Engine), L/h	2.10
4. Input Capacity, kg/h	1129

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

This test report should not be used for advertisement or for promotion of the product but only for evaluation purposes and should not be reproduced in whole or in part without AMTEC permission.